Nash Bargaining SQP+OptNet Layer: Mathematical Formulation

1 Decision Variables

For a family with M members, each member m has at most T activities. The per-activity decision vector is:

$$\mathbf{x}_{m,t} = \left[\, s_{m,t},\ e_{m,t},\ p^{\text{mode}}_{m,t,1}, \ldots, p^{\text{mode}}_{m,t,K},\ j_{m,t},\ d_{m,t} \,\right] \in \mathbb{R}^V, \tag{1}$$

where $V = 2 + K + 2 = 15$ (with $K = 11$ travel modes), and:

- $s_{m,t}$, $e_{m,t}$: start and end time in z-score normalization ($\mu = 12.946$ h, $\sigma = 4.691$ h),
- $p^{\text{mode}}_{m,t,k}$: probability of travel mode k, forming a simplex $\sum_{k=1}^{K} p^{\text{mode}}_{m,t,k} = 1$,
- $j_{m,t}$: joint travel probability ($\in [\varepsilon, 1-\varepsilon]$),
- $d_{m,t}$: driver probability ($\in [\varepsilon, 1-\varepsilon]$), where $\varepsilon = 10^{-3}$.

The full family decision vector is obtained by concatenating all per-member, per-activity vectors:

$$\mathbf{x} = \left[\, \mathbf{x}_{1,1}, \ldots, \mathbf{x}_{1,T}, \ldots, \mathbf{x}_{M,1}, \ldots, \mathbf{x}_{M,T} \,\right] \in \mathbb{R}^n, \quad n = M \times T \times V. \tag{2}$$

2 Utility Function

The utility of member m is defined as:

$$\begin{aligned} U_m(\mathbf{x}) = \beta_0 &+ \alpha_{\text{social}} \sum_{t=1}^{T} \tilde{o}_{m,t}\, a_{m,t} + \alpha_{\text{joint}} \sum_{t=1}^{T} j_{m,t}\, a_{m,t} \\ &+ \theta_{\text{escort}} \sum_{t=1}^{T} d_{m,t}\,(e_{m,t} - s_{m,t})\, a_{m,t}\, \mathbb{1}_{\text{adult}}(m) \\ &+ \sum_{t=1}^{T} \tilde{o}_{m,t} \left(\sum_{k=1}^{K} p^{\text{mode}}_{m,t,k}\, \theta_{\text{travel},k}\, \mathbb{E}[\mathrm{TT}_{m,t,k}] \right) a_{m,t}, \end{aligned} \tag{3}$$

where:

- β_0: learnable baseline utility,
- α_{social}: learnable social (out-of-home) preference weight,
- α_{joint}: learnable joint travel preference weight,
- θ_{escort}: learnable escort burden coefficient (typically negative),
- $\theta_{\text{travel},k}$: learnable mode-specific travel time disutility (vector of length K),
- $\tilde{o}_{m,t} = 1 - P(\text{dest}_{m,t} = \text{home})$: out-of-home probability from the destination head (fixed, not optimized),

- $\mathbb{E}[\mathrm{TT}_{m,t,k}]$: expected travel time for mode k at activity t, derived from the travel time matrix and destination probabilities (fixed),
- $a_{m,t}$: activity validity mask (1 if activity t of member m is valid, 0 otherwise),
- $\mathbb{1}_{\text{adult}}(m)$: indicator for adult members.

Lower bound. To ensure positivity for the log-utility in the Nash welfare function, a smooth lower bound is applied:

$$U_m \leftarrow U_m + \mathrm{softplus}(1 - U_m), \tag{4}$$

where $\mathrm{softplus}(z) = \log(1 + e^z)$. This guarantees $U_m \geq 1$ smoothly without gradient discontinuity. Utilities of invalid members are set to β_0 to avoid affecting the Nash objective.

3 Nash Welfare Objective

The Nash Bargaining solution maximizes the product of member utilities, equivalently the sum of log-utilities:

$$\max_{\mathbf{x}} \ W(\mathbf{x}) = \sum_{m=1}^{M} \log U_m(\mathbf{x}). \tag{5}$$

The objective $W(\mathbf{x})$ is **nonlinear**: the $\log(\cdot)$ composition is inherently nonlinear, and the escort term in U_m contains the bilinear product $d_{m,t}(e_{m,t} - s_{m,t})$.

4 Phase 1: Nonlinear Program

Phase 1 solves the full nonlinear program (NLP) to find a converged point $\mathbf{x}_{\text{conv}}$.

4.1 Optimization Problem

$$\max_{\mathbf{x}} \ W(\mathbf{x}) = \sum_{m=1}^{M} \log U_m(\mathbf{x}) \tag{6}$$

subject to the constraints in Table 1.

Table 1: Phase 1 constraints in variable space $\mathbf{x}$.

Constraint	Formulation	Type
(C1) Variable bounds	$x_i^{\text{lb}} \leq x_i \leq x_i^{\text{ub}}$	Linear
(C2) Time ordering	$e_{m,t} \geq s_{m,t}, \quad s_{m,t+1} \geq e_{m,t}$	Linear
(C3) Mode simplex	$\sum_{k=1}^{K} p_{m,t,k}^{\text{mode}} = 1$	Linear

The variable bounds (C1) are:

$$s_{m,t},\, e_{m,t} \in [z_{\min}, z_{\max}], \quad z_{\min} = \frac{1-\mu}{\sigma},\ z_{\max} = \frac{24-\mu}{\sigma}, \tag{7}$$

$$p_{m,t,k}^{\text{mode}} \in [\varepsilon, 1-\varepsilon], \tag{8}$$

$$j_{m,t},\, d_{m,t} \in [\varepsilon, 1-\varepsilon]. \tag{9}$$

Scope of Phase 1 constraints. Phase 1 handles only the simple per-variable and per-activity constraints (C1–C3) via projection. More complex family-level coordination constraints—vehicle capacity and joint travel consistency—involve nonlinear coupling between members and are *not* enforced during Phase 1. These constraints are introduced in Phase 2 (§5), where they are linearized at $\mathbf{x}_{\text{conv}}$ and incorporated as explicit linear constraints in the QP formulation.

4.2 SQP Algorithm

Phase 1 solves the NLP using Sequential Quadratic Programming (SQP) with BFGS inverse Hessian approximation. The constraints (C1–C3) are handled via projection after each step.

Algorithm 1 Phase 1: SQP with BFGS and Armijo Line Search

Require: Decoder output $\boldsymbol{\theta}$, masks, utility parameters
Ensure: Converged point $\mathbf{x}_{\text{conv}}$

$\mathbf{x} \leftarrow \boldsymbol{\theta}$ ▷ Initialize at decoder output
$H \leftarrow I$ ▷ Initial inverse Hessian approximation
for $i = 0, \ldots, I_{\max} - 1$ **do**
  $\mathbf{g} \leftarrow -\nabla_{\mathbf{x}} W(\mathbf{x})$ ▷ Negative gradient of Nash welfare
  if $\|\mathbf{g}\| < \epsilon_{\text{tol}}$ **then break** ▷ Converged
  end if
  if $i > 0$ **then**
    $\mathbf{s} \leftarrow \mathbf{x}^{(i)} - \mathbf{x}^{(i-1)}, \quad \mathbf{y} \leftarrow \mathbf{g}^{(i)} - \mathbf{g}^{(i-1)}$
    $H \leftarrow \text{BFGS-}\textsc{Update}(H, \mathbf{s}, \mathbf{y})$ ▷ With Powell's damping
  end if
  $\mathbf{p} \leftarrow -(H + \delta I)\,\mathbf{g}$ ▷ Search direction; $\delta = 10^{-3}$
  $\alpha \leftarrow \textsc{Armijo-Search}(\mathbf{x}, \mathbf{p})$ ▷ Backtracking line search
  $\mathbf{x} \leftarrow \textsc{Project}(\mathbf{x} + \alpha\,\mathbf{p})$ ▷ Project onto (C1)–(C3)
end for
return $\mathbf{x}_{\text{conv}} \leftarrow \mathbf{x}$

BFGS inverse Hessian update with Powell's damping. The matrix H approximates $[\nabla^2(-W)]^{-1}$, updated via the damped BFGS formula. Given step $\mathbf{s} = \mathbf{x}^{(i)} - \mathbf{x}^{(i-1)}$ and gradient change $\mathbf{y} = \mathbf{g}^{(i)} - \mathbf{g}^{(i-1)}$:

1. If $\mathbf{s}^\top \mathbf{y} < 0.2\,\mathbf{s}^\top H \mathbf{s}$, apply Powell's damping:

$$\phi = \frac{0.8\,\mathbf{s}^\top H \mathbf{s}}{\mathbf{s}^\top H \mathbf{s} - \mathbf{s}^\top \mathbf{y}}, \quad \mathbf{y}_d = \phi\,\mathbf{y} + (1-\phi)\,H\mathbf{s}. \tag{10}$$

   Otherwise $\mathbf{y}_d = \mathbf{y}$.

2. Apply the BFGS update:

$$H^+ = \left(I - \rho\,\mathbf{s}\,\mathbf{y}_d^\top\right) H \left(I - \rho\,\mathbf{y}_d\,\mathbf{s}^\top\right) + \rho\,\mathbf{s}\,\mathbf{s}^\top, \quad \rho = \frac{1}{\mathbf{s}^\top \mathbf{y}_d}. \tag{11}$$

Armijo line search. Starting from $\alpha = 1$, backtrack with factor 0.5 until the sufficient decrease condition holds:

$$-W\big(\textsc{Project}(\mathbf{x} + \alpha\,\mathbf{p})\big) < -W(\mathbf{x}) - c\,\alpha\|\mathbf{p}\|^2, \tag{12}$$

where $c = 10^{-4}$ is the Armijo parameter.

Projection. The projection operator $\textsc{Project}(\cdot)$ enforces the linear constraints (C1)–(C3):

1. Clamp times: $s_{m,t}, e_{m,t} \in [z_{\min}, z_{\max}]$.
2. Enforce temporal ordering: for $t = 1, \ldots, T$ sequentially, set $s_{m,t} \geq e_{m,t-1}$ and $e_{m,t} \geq s_{m,t}$.
3. Project mode probabilities onto the simplex: clamp to $[\varepsilon, \infty)$ and renormalize.
4. Clamp $j_{m,t}, d_{m,t} \in [\varepsilon, 1 - \varepsilon]$.

5 Phase 2: Quadratic Program

Phase 2 constructs a QP around the converged point $\mathbf{x}_{\text{conv}}$ from Phase 1. The displacement variable is $\mathbf{d} = \mathbf{x} - \mathbf{x}_{\text{conv}}$. The QP enforces two groups of constraints: (i) the Phase 1 constraints (C1–C3) re-expressed in the displacement space, and (ii) **new** family-level coordination constraints (C4–C5) that are introduced for the first time in Phase 2.

5.1 QP Formulation

$$\min_{\mathbf{d}} \ \frac{1}{2}\mathbf{d}^\top Q \mathbf{d} + \mathbf{p}^\top \mathbf{d} \quad \text{s.t.} \quad G\mathbf{d} \leq \mathbf{h}, \quad A\mathbf{d} = \mathbf{b}. \tag{13}$$

Quadratic term Q. Q is a diagonal matrix encoding the Nash Hessian approximation plus anchor regularization:

$$Q = \operatorname{diag}\left(\frac{1}{U_m^2} + \lambda\right)_i, \tag{14}$$

where U_m is the utility of member m evaluated at $\mathbf{x}_{\text{conv}}$ (Eq. 3), the index i maps to member m via the variable layout, and λ is the fixed anchor weight. The term $1/U_m^2$ approximates the diagonal of the Nash welfare Hessian $\nabla^2 W$, and λ penalizes deviation from the decoder output $\boldsymbol{\theta}$.

Linear term p.

$$\mathbf{p} = \lambda\left(\mathbf{x}_{\text{conv}} - \boldsymbol{\theta}\right), \tag{15}$$

where $\boldsymbol{\theta}$ is the decoder output. This term anchors the QP solution toward $\boldsymbol{\theta}$.

Nash correction weight. The relative influence of Nash coordination versus decoder fidelity is:

$$\gamma_{\text{Nash}} = \frac{1/U_m^2}{1/U_m^2 + \lambda}. \tag{16}$$

5.2 Inequality Constraints: $G\mathbf{d} \leq \mathbf{h}$

Inherited from Phase 1 (re-expressed in d)

(C1′) Variable bounds. Each variable must stay within its feasible range:

$$-d_i \leq x_i^{\text{conv}} - x_i^{\text{lb}}, \quad d_i \leq x_i^{\text{ub}} - x_i^{\text{conv}}. \tag{17}$$

This yields $2n$ inequality rows.

(C2′) Time ordering. The temporal constraints $e_{m,t} \geq s_{m,t}$ and $s_{m,t+1} \geq e_{m,t}$ become:

$$d_{s_{m,t}} - d_{e_{m,t}} \leq e_{m,t}^{\text{conv}} - s_{m,t}^{\text{conv}}, \tag{18}$$

$$d_{e_{m,t}} - d_{s_{m,t+1}} \leq s_{m,t+1}^{\text{conv}} - e_{m,t}^{\text{conv}}. \tag{19}$$

These remain linear, as in Phase 1.

New family-level coordination constraints (introduced in Phase 2)

(C4) Vehicle capacity. This constraint is introduced for the first time in Phase 2. The vehicle capacity constraint $\sum_{m,t} d_{m,t} \cdot p^{\text{car}}_{m,t} \cdot \mathbb{1}_{\text{adult}}(m) \cdot a_{m,t} \leq n_{\text{veh}}$ involves the bilinear product $d_{m,t} \cdot p^{\text{car}}_{m,t}$ (both are decision variables), which cannot be directly included in a QP. It is linearized via first-order Taylor expansion around $\mathbf{x}_{\text{conv}}$:

$$\sum_{m,t} \left(p^{\text{car},0}_{m,t} \, \Delta d_{m,t} + d^0_{m,t} \, \Delta p^{\text{car}}_{m,t} \right) \mathbb{1}_{\text{adult}}(m) \, a_{m,t} \leq n_{\text{veh}} - \sum_{m,t} d^0_{m,t} \, p^{\text{car},0}_{m,t} \, \mathbb{1}_{\text{adult}}(m) \, a_{m,t}, \tag{20}$$

where superscript 0 denotes values at $\mathbf{x}_{\text{conv}}$, and Δ denotes the displacement component. This linearization is valid locally around $\mathbf{x}_{\text{conv}}$, which is why this constraint is enforced in Phase 2 (after convergence) rather than Phase 1. It contributes a single linear inequality row.

5.3 Equality Constraints: $A\mathbf{d} = \mathbf{b}$

Inherited from Phase 1 (re-expressed in d)

(C3′) Mode probability normalization. Since $\sum_k p^{\text{mode}}_{m,t,k} = 1$ holds at $\mathbf{x}_{\text{conv}}$, the displacement must preserve this:

$$\sum_{k=1}^{K} \Delta p^{\text{mode}}_{m,t,k} = 0, \quad \forall\, m, t. \tag{21}$$

New family-level coordination constraint (introduced in Phase 2)

(C5) Joint travel consistency. This constraint is introduced for the first time in Phase 2. When two members m_1, m_2 are both engaged in joint travel at activity t, their activity attributes should agree. The activation is determined by a sigmoid gate evaluated at $\mathbf{x}_{\text{conv}}$ and fixed as a binary decision:

$$w_{m_1,m_2,t} = \mathbb{1}\!\left[\sigma\!\left(10(j^0_{m_1,t} \cdot j^0_{m_2,t} - \tau)\right) > 0.1\right], \tag{22}$$

where $\tau = 0.5$ is the joint consistency threshold, $\sigma(\cdot)$ is the sigmoid function, and superscript 0 denotes values at $\mathbf{x}_{\text{conv}}$. The sigmoid of the bilinear product $j_{m_1,t} \cdot j_{m_2,t}$ is inherently nonlinear; evaluating it at $\mathbf{x}_{\text{conv}}$ and fixing the result as a binary gate converts this into a fixed set of linear equalities suitable for the QP.

For each active pair ($w_{m_1,m_2,t} = 1$):

$$\Delta s_{m_1,t} - \Delta s_{m_2,t} = s^0_{m_2,t} - s^0_{m_1,t}, \tag{23}$$

$$\Delta e_{m_1,t} - \Delta e_{m_2,t} = e^0_{m_2,t} - e^0_{m_1,t}, \tag{24}$$

$$\Delta p^{\text{mode}}_{m_1,t,k} - \Delta p^{\text{mode}}_{m_2,t,k} = p^{\text{mode},0}_{m_2,t,k} - p^{\text{mode},0}_{m_1,t,k}, \quad \forall\, k, \tag{25}$$

$$\Delta j_{m_1,t} - \Delta j_{m_2,t} = j^0_{m_2,t} - j^0_{m_1,t}. \tag{26}$$

Each active pair contributes $2 + K + 1 = 14$ equality rows. Note that the right-hand sides drive the paired variables toward agreement: if the pair is active and the values differ at $\mathbf{x}_{\text{conv}}$, the QP will force them to converge.

5.4 Phase 1 → Phase 2 Connection

Table 2 summarizes the full constraint set in Phase 2.

Table 2: Phase 2 constraint summary. Constraints C1–C3 are inherited from Phase 1; C4–C5 are new in Phase 2.

Constraint	Phase 2 form	Origin	Linearization method
(C1′) Bounds	$G\mathbf{d} \leq \mathbf{h}$	Phase 1	Direct translation
(C2′) Time ordering	$G\mathbf{d} \leq \mathbf{h}$	Phase 1	Direct translation
(C3′) Mode simplex	$A\mathbf{d} = \mathbf{b}$	Phase 1	Conservation (zero-sum)
(C4) Vehicle capacity	$G\mathbf{d} \leq \mathbf{h}$	**New in Phase 2**	Taylor expansion at $\mathbf{x}_{\text{conv}}$
(C5) Joint consistency	$A\mathbf{d} = \mathbf{b}$	**New in Phase 2**	Sigmoid gate fixed at $\mathbf{x}_{\text{conv}}$

Algorithm 2 Phase 2: QP Solve with Gradient Re-attachment

Require: $\mathbf{x}_{\text{conv}}$, $\boldsymbol{\theta}$ (decoder output), masks, utility parameters
Ensure: $\mathbf{x}_{\text{final}}$ (with gradient to decoder and utility parameters)

— **Step 1: Construct QP parameters** —
$U_m \leftarrow$ Eq. (3) evaluated at $\mathbf{x}_{\text{conv}}$
$Q_{\text{diag}} \leftarrow 1/U_m^2 + \lambda$ (expanded per variable via member mapping) ▷ Eq. (14)
$\mathbf{p} \leftarrow \lambda(\mathbf{x}_{\text{conv}} - \boldsymbol{\theta})$ ▷ Eq. (15)
— **Step 2: Build constraints (C1′–C3′ inherited)** —
$(G, \mathbf{h}) \leftarrow$ Eqs. (17)–(20) ▷ C1′, C2′
$(A, \mathbf{b}) \leftarrow$ Eqs. (21)–(26) ▷ C3′
— **Step 3: Solve QP** —
$\mathbf{d}^* \leftarrow \arg\min_{\mathbf{d}} \frac{1}{2}\mathbf{d}^\top Q\mathbf{d} + \mathbf{p}^\top\mathbf{d}$ s.t. $G\mathbf{d} \leq \mathbf{h},\ A\mathbf{d} = \mathbf{b}$
— **Step 4: Re-attach gradient** —
$\mathbf{d}^* \leftarrow \textsc{DiagonalKKTBackward}(\mathbf{p}, Q_{\text{diag}}, \mathbf{d}^*)$ ▷ See §5.7
— **Step 5: Assemble final output** —
$\mathbf{x}_{\text{final}} \leftarrow \boldsymbol{\theta} + (\mathbf{x}_{\text{conv}} - \boldsymbol{\theta})_{\text{detach}} + \mathbf{d}^*$ ▷ Eq. (27)
$\mathbf{x}_{\text{final}} \leftarrow \textsc{Project}(\mathbf{x}_{\text{final}})$
return $\mathbf{x}_{\text{final}}$

5.5 QP Solve

5.6 Final Solution Assembly

The final output is composed as:

$$\mathbf{x}_{\text{final}} = \boldsymbol{\theta} + \underbrace{(\mathbf{x}_{\text{conv}} - \boldsymbol{\theta})}_{\text{no gradient}} + \underbrace{\mathbf{d}^*(\boldsymbol{\theta}, Q)}_{\text{carries gradient}}, \tag{27}$$

where the SQP displacement $(\mathbf{x}_{\text{conv}} - \boldsymbol{\theta})$ is treated as a constant (no gradient flows through Phase 1), and $\mathbf{d}^*$ depends on $\boldsymbol{\theta}$ through $\mathbf{p}$ (Eq. 15) and on utility parameters through Q (Eq. 14).

5.7 Diagonal KKT Backward

For the diagonal QP $\min_{\mathbf{d}} \frac{1}{2}\mathbf{d}^\top \operatorname{diag}(Q_{\text{diag}})\,\mathbf{d} + \mathbf{p}^\top\mathbf{d}$, the unconstrained closed-form solution is $\mathbf{d}^* = -\mathbf{p}/Q_{\text{diag}}$. We use this diagonal structure to approximate the gradients of the constrained solution:

$$\frac{\partial d_i^*}{\partial p_i} \approx -\frac{1}{Q_{\text{diag},i}}, \qquad \frac{\partial d_i^*}{\partial Q_{\text{diag},i}} \approx -\frac{d_i^*}{Q_{\text{diag},i}}. \tag{28}$$

Given upstream gradient $\bar{\mathbf{d}}$ from the loss function, the gradients propagate as:

$$\bar{\mathbf{p}} = -\bar{\mathbf{d}} \,/\, Q_{\text{diag}} \qquad \longrightarrow \text{gradient to } \boldsymbol{\theta} \text{ (decoder)}, \tag{29}$$

$$\bar{Q}_{\text{diag}} = -\bar{\mathbf{d}} \odot \mathbf{d}^* \,/\, Q_{\text{diag}} \qquad \longrightarrow \text{gradient to utility parameters via } 1/U_m^2. \tag{30}$$

6 Straight-Through Log Unpack

After optimization, probability variables (p^{mode}, j, d) must be converted back to log-probabilities for the downstream cross-entropy loss. A naive $\log(p)$ introduces a $1/p$ factor in the gradient, causing instability for small probabilities.

The straight-through log estimator is applied:

$$\tilde{p}_k = p_k + \big(\log p_k - p_k\big)_{\text{detach}}, \tag{31}$$

so that the forward value equals $\log p_k$ while the backward gradient is the identity ($\partial \tilde{p}_k / \partial p_k = 1$). For binary heads (joint, driver), the scalar p is expanded to $[1 - p,\ p]$ before applying Eq. (31).

7 Learnable Parameters

Table 3 summarizes all parameters of the Nash Bargaining layer.

Table 3: Parameters of the Nash Bargaining layer.

Parameter	Dim	Initial Value	Description
β_0	1	2.0	Baseline utility
α_{social}	1	0.3	Out-of-home social preference
α_{joint}	1	0.1	Joint travel preference
θ_{escort}	1	-0.58	Escort burden (adults)
θ_{travel}	K=11	$[-1.5, -0.4, \ldots]$	Mode-specific travel time disutility
λ (fixed)	1	0.3	Anchor weight (not learned)

All learnable parameters are updated via backpropagation through the diagonal KKT backward (Eqs. 29–30). The anchor weight λ is a fixed hyperparameter controlling the trade-off between Nash coordination and fidelity to the decoder output.

8 Two-Phase Flow Summary

1. **Pack**: Convert decoder predictions (logits) to probability-space vector $\boldsymbol{\theta}$.
2. **Phase 1 (SQP)**: Starting from $\mathbf{x}_0 = \boldsymbol{\theta}$, iterate BFGS + Armijo to find $\mathbf{x}_{\text{conv}}$ (Alg. 1). Handles per-variable constraints (C1–C3) via projection.
3. **Phase 2 (QP)**: Construct QP (Eq. 13) using $\mathbf{x}_{\text{conv}}$ and $\boldsymbol{\theta}$. Enforces all constraints: (C1′–C3′) inherited from Phase 1, plus **new** family-level coordination constraints (C4) vehicle capacity and (C5) joint consistency, linearized at $\mathbf{x}_{\text{conv}}$. Gradient flows via diagonal KKT backward (Alg. 2).
4. **Project**: Enforce feasibility on $\mathbf{x}_{\text{final}}$.
5. **Unpack**: Convert back to prediction dict using straight-through log (Eq. 31).

Gradient flows from loss $\to$ unpack $\to$ $\mathbf{x}_{\text{final}}$ $\to$ $\mathbf{d}^*$ $\to$

- $\mathbf{p} \to \boldsymbol{\theta} \to$ decoder parameters,
- $Q_{\text{diag}} \to 1/U_m^2 \to$ learnable utility parameters.